

Sana Al Hamimidi

San Jose, CA 669-210-5331 s.althamimidi04@gmail.com [linkedin.com/in/sana-al-hamimidi](https://www.linkedin.com/in/sana-al-hamimidi)

EDUCATION

San José State University San Jose, CA

B.S. Data Science **GPA: 3.5**

May 2026

Relevant Coursework: Statistics, Data Modeling, Processing Big Data, Data Structures & Algorithms, Machine Learning, Bioinformatics, SQL, R, Python

TECHNICAL SKILLS

Languages	Python, SQL, R, Java
Libraries & ML	scikit-learn, XGBoost, pandas, matplotlib, seaborn, plotly
Tools & Platforms	Git, Excel, Tkinter, Jupyter, Google Cloud (Cloud Run, App Engine, GCS), Docker, Flask
Databases	MySQL

PROJECTS

Vehicle Financial Decision Engine

Jan 2026 – May 2026

CS163 Senior Capstone · San José State University · Team of 2 | Live: cs163seniorproject.uw.r.appspot.com

- ▶ Built an end-to-end ML pipeline predicting used car resale prices from 50,000+ Craigslist listings and 3.6M California DMV records; XGBoost model achieved $R^2 = 0.779$, RMSE = \$5,891.
- ▶ Engineered a demand score feature by mapping ZIP codes to California regions and aggregating DMV registrations (2019–2024) by region, make, and fuel type; fuel type was the top predictor at 27% importance.
- ▶ Deployed a Flask REST API (Docker → Google Cloud Run) with model artifacts on GCS and static site on App Engine; uncovered a \$28,550 regional price gap for the same vehicle across California markets.

Medicaid Drug Utilization Analysis

Jan 2026 – Apr 2026

CS133 Data Visualization · Team of 3

- ▶ Analyzed 2.6M rows from the 2025 CMS State Drug Utilization Dataset; built interactive Plotly visualizations revealing BIKTARVY (\$3.1B) as the top-cost drug and GLP-1 drugs accounting for 17.2% of total Medicaid reimbursements across 21 states.
- ▶ Trained Random Forest, Logistic Regression, and Decision Tree classifiers to predict high vs. low Medicaid GLP-1 coverage (target: >80% coverage rate); Random Forest achieved 95.5% accuracy, ROC-AUC 0.99, and macro-F1 0.89 using `class_weight='balanced'` to address 90/10 class imbalance.
- ▶ Applied ColumnTransformer pipeline with StandardScaler on numerical features and OneHotEncoder on 4 categorical features; found state-level indicators outweighed drug identity as the dominant coverage predictor.

Human Activity Recognition (KU-HAR)

Oct 2025 – Dec 2025

Machine Learning Course Project · Team of 5

- ▶ Built a full HAR pipeline on 20,750 samples from 90 participants across 18 activities using smartphone accelerometer and gyroscope data; extracted 24 time-domain features per 3-second window after testing 10 window sizes.
- ▶ Benchmarked 6 classifiers (Random Forest, XGBoost, SVM, Naive Bayes, AdaBoost, Decision Tree); XGBoost achieved the highest test accuracy at 94.77% with 0.9372 macro-F1, outperforming all others.

Online Cafe Storefront & Order Management System

Nov 2025 – Dec 2025

CS157a Database Systems · Team of 2

- ▶ Built a 3-tier order management system (Tkinter GUI + Python MySQL connector + MySQL) with a BCNF-normalized schema across 5 tables (Categories, Menu_Items, Customers, Orders, Order_Items) using foreign keys, UNIQUE constraints, and DECIMAL for precise financials.
- ▶ Implemented parameterized SQL queries and transactions for all CRUD operations; generated analytics reports surfacing peak revenue hour (13:00, \$277.82/hr) and top-5 spending customers.